

● MEDIUM

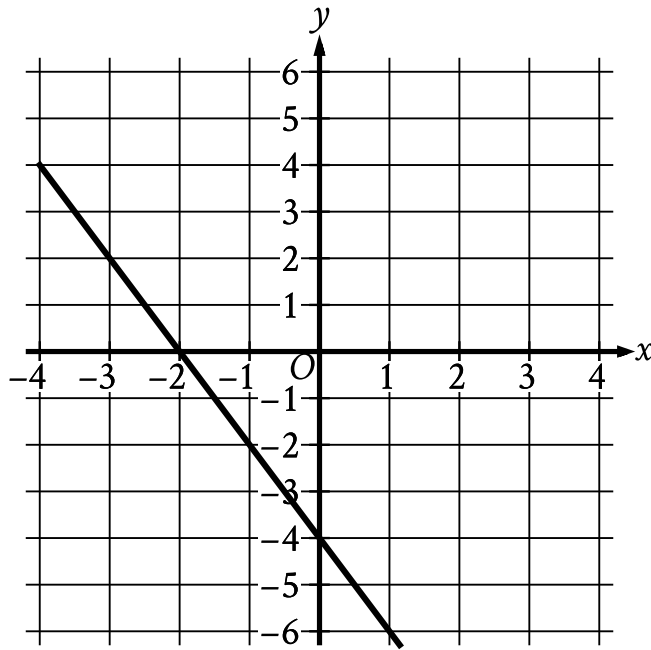
● ALGEBRA

Algebra

10 questions · ~15 min

17

The graph of $y = fx - 11$ is shown.



- The line slants sharply down from left to right.
- The line passes through the following points:
 - (negative 1 comma negative 2)
 - (0 comma negative 4)

Which equation defines the linear function f ?

- A. $fx = -13x - 11$
- B. $fx = -2x + 7$
- C. $fx = -13x + 7$
- D. $fx = -2x - 11$

Q2

ALGEBRA

$$y - 9x = 13$$

$$5x = 2y$$

What is the solution x, y to the given system of equations?

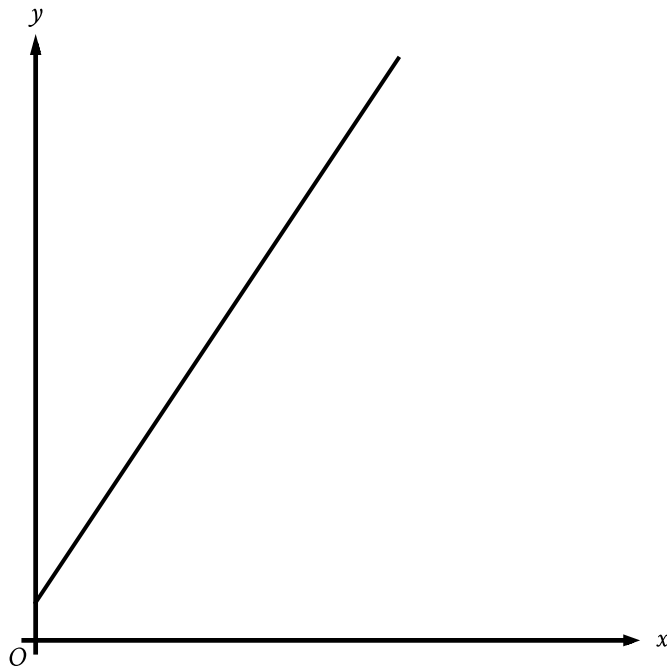
- A. $\frac{5}{2}, 1$
- B. $1, \frac{2}{5}$
- C. $-2, -5$
- D. $-5, -2$

Q3

ALGEBRA

Line t in the xy -plane has a slope of $-\frac{1}{3}$ and passes through the point $9, 10$. Which equation defines line t ?

- A. $y = 13x - \frac{1}{3}$
- B. $y = 9x + 10$
- C. $y = -\frac{x}{3} + 10$
- D. $y = -\frac{x}{3} + 13$



- The line begins on the y axis above the x axis.
- The line slants sharply up from left to right.

The graph represents the total charge, in dollars, by an electrician for x hours of work. The electrician charges a onetime fee plus an hourly rate. What is the best interpretation of the slope of the graph?

- A. The electrician's hourly rate
- B. The electrician's onetime fee
- C. The maximum amount that the electrician charges
- D. The total amount that the electrician charges

Q5

GRID-IN ALGEBRA

$$2a + 8b = 198$$

$$2a + 4b = 98$$

The solution to the given system of equations is a, b . What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A batch of banana milkshakes consists of 4 cups of ice cream and 2 bananas and has 1,114 milligrams mg of calcium. There is 276 mg of calcium in 1 cup of the ice cream used to make this batch of milkshakes. How much calcium, in mg, is in 1 banana?

- A. 5
- B. 10
- C. 419
- D. 1,104

Energy per Gram of Typical Macronutrients

Macronutrient	Food calories	Kilojoules
Protein	4.0	16.7
Fat	9.0	37.7
Carbohydrate	4.0	16.7

The table above gives the typical amounts of energy per gram, expressed in both food calories and kilojoules, of the three macronutrients in food. If x food calories is equivalent to k kilojoules, of the following, which best represents the relationship between x and k ?

- A. $k = 0.24x$
- B. $k = 4.2x$
- C. $x = 4.2k$
- D. $xk = 4.2$

$$\frac{1}{2}y = 4$$

$$x - \frac{1}{2}y = 2$$

The system of equations above has solution (x, y) . What is the value of x ?

A. 3

B. $\frac{7}{2}$

C. 4

D. 6

Q9

ALGEBRA

Line r is defined by the equation $4x - 9y = 3$. Line s is parallel to line r in the xy -plane. What is the slope of line s ?

- A. $\frac{9}{4}$
- B. $\frac{4}{9}$
- C. -4
- D. -9

Q10

ALGEBRA

The function f is defined by $fx = \frac{x+15}{5}$, and $fa = 10$, where a is a constant. What is the value of a ?

- A. 5
- B. 10
- C. 35
- D. 65